

Alice in Subtractionland

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Alice and Bob are playing the so-called “subtraction game.” In this game, there are n tokens in a single pile, and each player, starting with Alice, takes turns removing between 1 and k tokens from the top of the pile. The goal is to be the last person to pick up a token.

(a) Given any n and k , determine which player wins with optimal play, and describe a simple algorithm by which to make the optimal move every time.

(b) Suppose there are instead two piles, perhaps with differing numbers of tokens, and Alice and Bob may only remove tokens from one pile during their turn. Who wins then?

(c) Now, suppose each token has a weight $w \leq k$ less than k . The players are allowed to remove some subset of tokens from the top of the pile whose weights have a sum that is at most k . Describe an algorithm for optimal play. Is the outcome of the game still determined solely by the choice of n and k ?

The subtraction game is one of many variants of the game Nim (confusingly, the subtraction game itself is also sometimes referred to as “Nim”). Nim itself involves multiple piles of tokens, and the players are allowed to remove any number of tokens from each pile, with the same goal of being the last player to make a move.

Interestingly, it can be shown that the subtraction game and the presented variants are equivalent to a game of Nim, for appropriately chosen values of p and $n_1, n_2, \dots, n_p$; this is a result of the [Sprague-Grundy theorem](#).